



BRAC UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Final Exam, Summer 2024

Course: CSE437 (Data Science: Coding With Real-World Data)
Full Marks: 40
Time: 24 hours

1. Consider the following dataset, **apply** K-Medoids clustering algorithm to find out two clusters. Afterward, **assign** cluster names as a Zoologists. [CO2, C3, Mark: 8 + 2]

Subjects	Number of Legs	Number of Tails
S1	quadrupedal	1
S2	bipedal	1
S3	bipedal	0
S4	quadrupedal	0

2. Consider the following dataset of 3 features namely X, Y, and Z. **Determine** the principal components and select the necessary components so that the transformed dataset holds 95% of the normalized cumulative variances. [CO2, C3, Marks: 8 + 2]

	X	Y	Z
Instance 1:	-2	0	-2
Instance 2:	0	2	2
Instance 3:	2	-2	0

Note that you have to transform the eigenvectors to unit vectors before forming the feature vector i.e., you have to determine the eigenvectors

$\begin{pmatrix} x_i \\ x_{i+1} \\ \vdots \\ x_n \end{pmatrix}$ and afterward, each of the eigenvectors must be scaled to unit length

eigenvector $\begin{pmatrix} x'_i \\ x'_{i+1} \\ \vdots \\ x'_n \end{pmatrix}$ using the equation $x'_i = \frac{x_i}{\|x\|_2}$, Where, $\|x\|_2$ is the l_2

norm of vector $x = [x_i, x_{i+1}, \dots, x_n]$.

3. **Find** the association rules using the Apriori algorithm considering the following transactions of a smartphone center. You have to show all the necessary computations to find out the association rules. You have to determine association rules of each and every frequent itemsets of length greater than or equal to 2. [Assume that, min_support=0.34 and min_confidence=0.6, min_lift=1.2] [CO3, C4, Marks: 10]

1	Smartpho	Charger	Earphone		
2	Smartpho	Charger			
3	Smartpho	Charger		Selfistick	Cover
4				Selfistick	Cover
5			Earphone		Cover
6	Smartphone			Selfistick	Cover

4. You think you are a genius. However, according to several statistics, only 2% of the whole population are real geniuses. Take the following IQ test 5 times. [CO3, C3, Marks: 10]
<https://iqtestfree.io/take-the-iq-test>
Is there sufficient evidence to support your claim considering the 5% significance level? You must attach your results (distribution graph only) of the 5 times IQ test.